Automated SOAR Playbook — Shuffle Endpoint Containment Workflow

This educational project demonstrates a cautious event-to-SOAR workflow. A Python utility normalizes a security event, applies a small demonstration rule, and can send the decision to a Shuffle HTTPS webhook. Any actual endpoint isolation must be configured as a separate, reviewed action inside Shuffle with strict policy gates.

Files

• `automated_soar_playbook.py` — event evaluator and optional Shuffle webhook notifier.

• `sample_security_event.json` — synthetic example event.

• `requirements.pdf` — hardware, software, functional, and safety requirements.

• `project_report.pdf` — report with the supplied Sandip University logo.

• `sandip_university_logo.png` — supplied logo asset.

Requirements

• Python 3.10 or newer

• No third-party Python packages required

• A Shuffle instance and a workflow webhook, if webhook notification is enabled

• An authorized test environment and a synthetic event source

Run locally

```
python automated_soar_playbook.py --input sample_security_event.json
```

This writes `soar_decision.json`. By default the script only evaluates and records a decision; it does not contact Shuffle or isolate an endpoint.

Notify a Shuffle workflow

Configure a Shuffle webhook that accepts the event decision and includes validation, endpoint allowlisting, logging, and human approval for disruptive actions. Keep the webhook URL secret.

Windows PowerShell

```
$env:SHUFFLE_WEBHOOK_URL = "https://your-shuffle-host.example/webhook/your-workflow-id"
python automated_soar_playbook.py --input sample_security_event.json --notify
```

macOS/Linux

```
export SHUFFLE_WEBHOOK_URL='https://your-shuffle-host.example/webhook/your-workflow-id'
python automated_soar_playbook.py --input sample_security_event.json --notify
```

Use HTTPS. Never commit webhook URLs, API tokens, or credentials to GitHub. The webhook URL format depends on your Shuffle deployment.

Recommended Shuffle playbook gates

1. Validate event schema and source authenticity.

2. Deduplicate events and correlate endpoint identity.

3. Check endpoint against an explicit lab allowlist; never contain critical/shared infrastructure automatically.

4. Apply severity and confidence thresholds.

5. Require human approval for disruptive containment unless a formally approved policy says otherwise.

6. Execute endpoint isolation only through an authorized EDR/endpoint integration.

7. Record the action, result, and operator/workflow identity.

8. Provide a tested rollback/reconnect procedure and alert the incident-response team.

Important limitations

• This script does not directly isolate endpoints.

• The demonstration rule is not a malware verdict and must be adapted to validated signals.

• A successful webhook request only means the endpoint accepted the HTTP request; it does not prove the workflow completed.

• Shuffle workflow nodes, credentials, and EDR integrations must be configured separately.

• Use only systems you own or are explicitly authorized to administer.

License

Educational project template. Add an appropriate license before redistribution.